The M/G/1 Queue

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Abstract

Residual service time and the Pollaczek–Khinchin formula, the $M/M/1$ and $M/D/1$ special cases, and $M/G/1$ with vacations.

1 Pollaczek-Khinchin formula

The Pollaczek-Khinchin formula is given as

$$\bar{W} = \frac{\lambda \left\langle X^2 \right\rangle}{2(1-\rho)}$$

where $\bar{W}$ is expected waiting time,$\left\langle X^2 \right\rangle = \left\langle X^2 \right\rangle_{p(X)}$ is the second moment under the service time distribution, the utilisation factor is given as $\rho = \lambda \bar{X}$ and $\bar{X} = \langle X \rangle_{p(X)}$ is the mean under the service time distribution.

2 Proof of Pollaczek-Khinchin Formula

In the examples, we review the first part of a simple proof of the Pollaczek-Khinchin formula. In our proof we will make three assumptions:

1. The queue is FIFO queue, *i.e.*customers are served in the order they arrive. This assumption may be relaxed but it will simplify our proof.
2. The arrival times are statistically independent of the service times.
3. The arrival time distribution has a variance which is finite.

To formulate the proof we introduce a concept known as the mean residual service time, $\bar{R}$. We show that the average waiting time is related to the mean residual service time as follows:

$$\bar{W} = \frac{\bar{R}}{1-\rho}$$

We now have the average queueing time in terms of the mean residual service time. To complete the Pollaczek-Khinchin formula we simply need to calculate $\bar{R}$.

3 Residual Service Time from Departures Perspective

We again turn to a graphical proof to obtain the value of the mean residual service time. In Figure 1 we present a plot of the residual service time over time.



Figure 1: Graphical computation of mean residual time. $M(t)$ is the number of service completions in the interval $[0, t]$.

The time average of the mean residual service time can then be calculated in terms of the area under the curve, $A(t)$, which is given by the area under each of the triangles,

$$A(t) = \frac{1}{2}\sum_{i=1}^{M(t)} X_i^2.$$

The average service time is the area divided by time, t,

$$\begin{aligned}\frac{A(t)}{t} &= \frac{M(t)}{t}\frac{A(t)}{M(t)} \\ &= \frac{1}{2}\frac{M(t)}{t}\frac{\sum_{i=1}^{M(t)} X_i^2}{M(t)}.\end{aligned}$$

where we have introduced the number of arrivals after time t, as $M(t)$. As $t \to \infty$, if the variance of the service time distribution is finite, we find that

$$\bar{R} = \frac{1}{2}\lambda\left\langle X^2\right\rangle.$$

Again, assuming that time averages and ensemble averages are equal, we substitute back into (2) to obtain the Pollaczek-Khinchin formula

$$\bar{W} = \frac{\lambda\left\langle X^2\right\rangle}{2(1-\rho)}.$$

This implies that the total waiting time in the system is

$$\bar{T} = \bar{X} + \frac{\lambda\left\langle X^2\right\rangle}{2(1-\rho)}$$

and through Little's formula we can obtain the expected queue length, $\bar{N}_Q$, and the expected number in the system, $\bar{N}$. **Note that derivation of this formula was Question 1 in the 2002-2003 paper**. Now consider Examples 2, 3 and 4.3 at the end of this chapter for the use of the Pollaczek-Khinchin formula.

3.1 Table of Delays and Average Numbers in System

In the table below we summarise average waiting times and various other characteristics of the systems for the Pollaczek-Khinchin formula.

Expected value	General	Exponential	Deterministic
$\bar{W}$	$\frac{\lambda}{2(1-\rho)}\left\langle X^2\right\rangle$	$\frac{\rho}{\mu(1-\rho)}$	$\frac{\rho}{2\mu(1-\rho)}$
$\bar{T}=\bar{W}+\frac{1}{\mu}$	$\frac{\lambda}{2(1-\rho)}\left\langle X^2\right\rangle+\frac{1}{\mu}$	$\frac{1}{\mu(1-\rho)}$	$\frac{2-\rho}{2\mu(1-\rho)}$
$\bar{N}_Q=\lambda\bar{W}$	$\frac{\lambda^2}{2(1-\rho)}\left\langle X^2\right\rangle$	$\frac{\rho^2}{(1-\rho)}$	$\frac{\rho^2}{2(1-\rho)}$
$\bar{N}=\lambda\bar{W}+\rho=$	$\frac{\lambda^2}{2(1-\rho)}\left\langle X^2\right\rangle+\rho$	$\frac{\rho}{1-\rho}$	$\frac{\rho(2-\rho)}{2(1-\rho)}$

4 Automatic Repeat Request (ARQ) System.

The $M/G/1$ queue system can be used for modelling various systems of interest. In an ARQ system packets are transmitted and the system waits for an acknowledgement. If no acknowledgement is received the system retransmits. In this section we review Example 3.15 from [Bertsekas and Gallager, 1992]. A schematic of the system is given in Figure 2. For the purposes of this analysis, we will assume

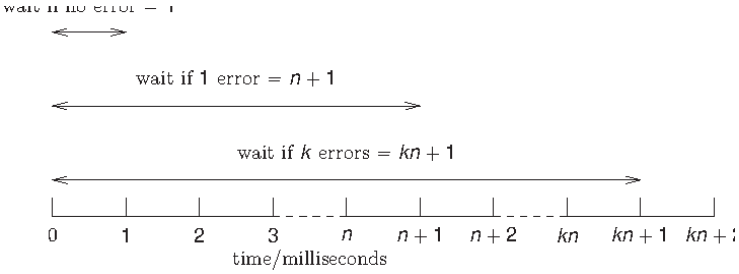


Figure 2: The ARQ System.

1. That each packet takes 1 millisecond to be transmitted.
2. The system waits a maximum of $n-1$ milliseconds for an acknowledgement. If there is no acknowledgement the system retransmits.
3. The acknowledgement occurs if the packet is received without errors. Otherwise acknowledgement doesn't occur.
4. The probability of an error in a packet is taken to be p and we take errors to occur independently.
5. That packets arrive as a Poisson process with rate λ.

In practice, the fourth assumption (independence of errors) is probably unrealistic . If an error occurs due to problems in the transmission channel, it is likely that several packets in a row will not be received, so the errors do not occur independently. The time interval between transmission and

re-transmission will be $1 + kn$ milliseconds with probability $(1-p)\,p^k$. This time interval is associated with k retransmissions and one successful transmission. The probability of this being the service time is given by

$$P\left(X = 1 + kn\right) = (1-p)\,p^k$$

which is recognised as a *geometric distribution.*

4.1 Average Service Time

Using this distribution we can compute the expected service time which is given by

$$\begin{aligned}
\langle X \rangle = & \sum_{k=0}^{\infty} (1+kn)\,(1-p)\,p^k \\
= & \quad (1-p)\left(\sum_{k=0}^{\infty} p^k + n\sum_{k=0}^{\infty} kp^k\right) \\
= & \quad 1 + n\frac{p}{1-p}
\end{aligned}$$

since $\sum_{k=0}^{\infty} p^k = \frac{1}{1-p}$ and $\sum_{k=0}^{\infty} kp^k = \frac{p}{(1-p)^2}$.

4.2 ARQ Utilization Factor

Strictly speaking, it probably doesn't make sense to use the term `utilization factor' for the ARQ system. This is because during a large portion of the wait it is not actually`utilizing' the channel, it is merely waiting for an acknowledgement. However we can still consider, $\rho = \lambda\,\langle X \rangle$, to be the proportion of time that the system will be `busy'. This`busy factor',

$$\rho = \lambda + \lambda n\frac{p}{1-p},$$

obviously still has to be less than 1, otherwise the steady state would not exist. This busy time is affected by three parameters: the rate of arrivals, λ, the amount of time that the system waits for an acknowledgement, n, and the probability of an error, p. Note that as $p \to 1$ the second term $\to \infty$.

4.3 ARQ Second Moment

The second moment of the service time is

$$\begin{aligned}\left\langle X^2 \right\rangle &= \sum_{k=0}^{\infty} (1+kn)^2 (1-p) p^k \\ &= (1-p)\left(\sum_{k=0}^{\infty} p^k + 2n\sum_{k=0}^{\infty} kp^k + n^2 \sum_{k=0}^{\infty} k^2 p^k\right) \\ &= 1 + \frac{2np}{1-p} + \frac{n^2\left(p+p^2\right)}{(1-p)^2}\end{aligned}$$

since

$$\sum_{k=0}^{\infty} k^2 p^k = \frac{p+p^2}{(1-p)^3}$$

These moments can be used with the Pollaczek-Khinchin formula to obtain statistics for the queue. Examples

1. The Residual Service Time The residual service time is the amount of service time that remains for the customer under service from the perspective of a customer who has just arrived in the queue. The first part of the proof occurs from the perspective of the arriving customer. The arriving customer is considered to be the ith customer. In the table below we introduce some simple definitions of terms.

W_i	=	Waiting time in queue of ith customer.
R_i	=	'Residual service time' seen by the ith customer.
X_i	=	Service time of the ith customer.
k_i	=	Number of customers in the system when the ith customer arrives.

2. **Markov Service times**. Compute the average waiting time associated with exponentially distributed service times using the Pollaczek-Khinchin formula.
3. **Deterministic Service times ($M/D/1$)**. Some systems have deterministic service times. For example, transmission of fixed length packets over a channel with a fixed bandwidth.
4. **Uniform Service Times** ($M/U/1$). Consider a uniform probability distribution is defined between 0 and b for $X > 0$ as

$$U(X|b) = \begin{cases} \frac{1}{b} & 0 < X \leq b \\ 0 & X > b \end{cases}$$

show that the mean of this uniform distribution is given by $\frac{b}{2}$ and its variance by $\frac{b^2}{12}$ and find the expected waiting time for an $M/U/1$ queue. 5. $M/G/1$ Queues with Vacations Consider a server which, when it is free, attends to some other business. For example it could transmit control and record keeping traffic. These periods are sometimes known as vacations (which is an American word that means holidays). The vacation periods are characterised by 6. If a new arrival arrives during a vacation it must wait until the vacation is over before being dealt with. 7. If a vacation ends and the server is idle it immediately goes on a new vacation.

We denote the length of these successive vacations to be V_{1}, V_{2}, $\dots$. We assume t

8. $M/G/1$**with Vacations**. A bar has Poisson arrivals and one member of staff serving. When there are no customers at the bar, the staff member leaves the bar to check on the kitchen. Assuming these checks are exponentially distributed in length, with mean $\bar{V}$ and that customer service times are exponentially distributed with mean $\bar{X}$ what is the average waiting time in the queue?}

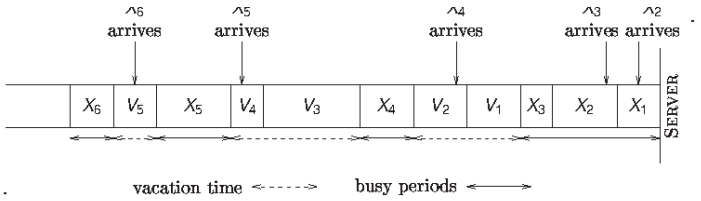


Figure 3: Schematic of $M/G/1$ queue

4.4

*{Answer} Given the notation we can draw a simple schematic of the system as it is perceived by the ith arrival. This is shown in Figure 4.4.



Figure 4: residualServiceOverview

We can use the figure to immediately write down that the wait for the i th customer will be given by

$$W_i = R_i + \sum_{j=i-(k_i-1)}^{i-1} X_j$$

summing over both sides (i.e. for all the arriving customers up to the I th) we obtain

$$\sum_{i=1}^{I} W_i = \sum_{i=1}^{I} R_i + \sum_{i=1}^{I} \sum_{j=i-(k_i-1)}^{i-1} X_j$$

dividing through by I gives us

$$\sum_{i=1}^{I} \frac{W_i}{I} = \sum_{i=1}^{I} \frac{R_i}{I} + \sum_{i=1}^{I} \sum_{j=i-(k_i-1)}^{i-1} \frac{X_j}{I}.$$

In the limit as $I \to \infty$ this becomes

$$\lim_{I\to\infty} \sum_{i=1}^{I} \frac{W_i}{I} = \lim_{I\to\infty} \sum_{i=1}^{I} \frac{R_i}{I} + \lim_{I\to\infty} \sum_{i=1}^{I} \frac{(k_i-1)X_i}{I},$$

the terms of which are recognised as time averages . If we assume time averages are equivalent to steady state averages this gives

$$\langle W \rangle = \langle R \rangle + \langle (k-1)\, X \rangle .$$

Now since the length of the queue as seen by the i th arrival is independent of the i th arrival's length (our second assumption) this means $\langle (k-1)X \rangle = \langle k-1 \rangle \langle X \rangle$ and since $(k-1)$ is the length of the queue we may write

$$\bar{W} = \bar{R} + \bar{N}_Q \bar{X}$$

Using Little's theorem for the queue we have

$$\bar{N}_Q = \lambda \bar{W}$$

which gives in turn

$$\bar{W} = \bar{R} + \lambda \bar{X} \bar{W}.$$

Using the fact that $\rho = \lambda \bar{X}$ (which stems from Little's formula for the packets under service) gives us

$$\bar{W} = \bar{R} + \rho \bar{W}$$

Leading to

$$\bar{W} = \frac{\bar{R}}{1-\rho}.$$

4.5

*{Answer} The exponentially distributed service times,

$$p(X) = \mu \exp(-\mu X),$$

have a mean of

$$\bar{X} = \frac{1}{\mu}$$

and a variance of $\frac{1}{\mu^2}$ leading to a second moment

$$\left\langle X^2 \right\rangle = \frac{2}{\mu^2}$$

Substituting into the Pollaczek-Khinchin formula we obtain

$$\begin{aligned} \bar{W} = & \quad \frac{\rho}{\mu(1-\rho)} \\ = & \quad \frac{\rho}{\mu - \lambda} \end{aligned}$$

which is identical to those we derived for the $M/M/1$ queue.

4.6

*{Answer} We represent a 'deterministic distribution' as

$$P(X) = \delta\left(X - \frac{1}{\mu}\right)$$

where $\delta(\cdot)$is the Dirac delta function. For this system

$$\langle X \rangle = \frac{1}{\mu}$$

and

$$\left\langle X^2 \right\rangle = \frac{1}{\mu^2}$$

i.e. the variance is zero. Substituting into the Pollaczek-Khinchin formula we obtain

$$\bar{W} = \frac{\rho}{2\mu(1-\rho)}.$$

- For simplicity we consider $\mu = \frac{1}{\langle X \rangle}$. So $\mu = \frac{2}{b}$ and thus $b = \frac{2}{\mu}$.
- So we can write the second moment in terms of μ. $\left\langle X^2 \right\rangle = \frac{b^2}{3}$.
- Substituting in $b = \frac{2}{\mu}$ we have $\left\langle X^2 \right\rangle = \frac{4}{3\mu^2}$.
- Substituting in to the Pollaczek-Khinchin formula we recover

$$\bar{W} = \frac{2\rho}{3\mu(1-\rho)}.$$

4.7

*{Answer} We need to compute residual service time For a system with vacations the residual service time is given by

$$\begin{aligned} \frac{1}{t}\int_0^t r(\tau)\,d\tau = & \quad \frac{1}{t}\sum_{i=1}^{M(t)} \frac{1}{2}X_i^2 + \frac{1}{t}\sum_{i=1}^{L(t)} \frac{1}{2}V_i^2 \\ = & \quad \frac{M(t)}{t}\frac{1}{2}\frac{\sum_{i=1}^{M(t)} X_i^2}{M(t)} + \frac{L(t)}{t}\frac{1}{2}\frac{\sum_{i=1}^{L(t)} V_i^2}{L(t)} \end{aligned}$$

where we have introduced the arrival process, $M(t)$, and this time we have also introduced a process which indicates how many vacations there have been, $L(t)$. This means we have to deal with the average number of vacations. On average the proportion of time available for vacations is $(1-\rho)$, i.e. that time where the transmission line is not in use. If the average length of a vacation at time t is $\bar{V}_t$ then the number of vacations that will have arrived at that time is

$$L(t) = \frac{t(1-\rho)}{\bar{V}_t}.$$

Taking the limit of (4.7) as $t \to \infty$ we can write

$$\bar{R} = \frac{1}{2}\lambda \left\langle X^2 \right\rangle + \frac{(1-\rho)\left\langle V^2 \right\rangle}{2\bar{V}}.$$

So given that the assumptions that we used to develop the formula

$$\bar{W} = \frac{\bar{R}}{1-\rho}$$

in Section 1 are still valid, we may write

$$\bar{W} = \frac{\lambda \left\langle X^2 \right\rangle}{2(1-\rho)} + \frac{\left\langle V^2 \right\rangle}{2\bar{V}}.$$

4.8

*{Answer} Exponential standard deviation = $\frac{1}{\mu}$, mean= $\frac{1}{\mu}$. So if mean is $\bar{V}$, standard deviation is $\bar{V}$ and variance is $\bar{V}^2$. Thus second moment is $\left\langle V^2 \right\rangle = 2\bar{V}^2$. Similarly for second moment of service time $\left\langle X^2 \right\rangle = 2\bar{X}^2$. Defining $\rho = \lambda\bar{X}$ we have

$$\bar{W} = \frac{\lambda\bar{X}}{(1-\rho)} + \bar{V} = \frac{\rho}{(1-\rho)} + \bar{V}.$$

Deterministic vacations would have lead to

$$\bar{W} = \frac{\lambda\bar{X}}{(1-\rho)} + \frac{\bar{V}}{2} = \frac{\rho}{(1-\rho)} + \frac{\bar{V}}{2}.$$

4.9 Thanks!

For more information on these subjects and more you might want to check the following resources.

- company: Trent AI
- book: The Atomic Human
- twitter: @lawrennd
- podcast: The Talking Machines
- newspaper: Guardian Profile Page
- blog: http://inverseprobability.com

4.10 Further Reading

- The $M/G/1$ queue, residual life, and vacations of Bertsekas and Gallager [1992]

4.11 Exercises

- Tutorial Sheet 3: triangular service time in Pollaczek–Khinchin of Lawrence [2006]

References

Dimitri P. Bertsekas and Robert G. Gallager. *Data Networks*. Prentice-Hall, Upper Saddle River, NJ, second edition, 1992. ISBN 0-132-00916-1.

Neil D. Lawrence. Erlang delay and loss and the M/G/1 queue. Network Performance Analysis Tutorial 3, University of Sheffield, 2006. URL https://mlatcl.github.io/npa/.